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Lukas Voss

Personal Statement

Driven by curiosity and a strong desire to learn, I excel in modeling systems and phenomena using numerical tools. With extensive experience in collaborating with international teams on strategic and high-impact quantitative projects across Europe and Asia, I combine a unique blend of training in both economics and theoretical physics. My passion for machine learning and my ability to unite and inspire teams enable me to thrive in dynamic environments.

Experience

07/23 – 09/23 **Fraunhofer IPA** *Cyber Cognitive Intelligence, Stuttgart, Germany* – Student Researcher
- Predict network traffic for component used in automated driving with Quantum Machine Learning (Python)

06/22 – 08/22 **BlackRock** *Portfolio Analytics Group, Budapest, Hungary* – Summer Analyst
- Built an evaluation pipeline to validate a new multi-asset multi-factor financial model by comparing its outputs to a previous model, enhancing data collection efficiency by 60% through infrastructure improvements
- Developed risk dashboard to automatically flag data inconsistencies between firm-wide and client databases across portfolios, using Python & SQL for data validation and integrated solution into a nightly update routine

03/22 – 06/22 **Roland Berger** *Restructuring, Performance, Transformation & Transaction, Munich, Germany* – Graduate Intern
- Prepared 15 investor proposals & intended deal structures for creditor committee for a distressed M&A target
- Interviewed 3 experts for due diligence, gaining insights into market fragmentation & competitor positioning

10/21 – 12/21 **KPMG** *Financial Risk & Treasury, Munich, Germany* – Intern
- Developed a treasury dashboard in PowerBI, integrating Python-based web scraping and Google News API to analyze real-time market data, including sentiment analysis (NLP) of news articles on 15 counterparties
- Presented project results to entire German Treasury practice (>60 FTE, including partners) in a nationwide call and gave a live demo to two clients, driving interest in advanced data-driven risk management solutions

Education (GPA out of 4.0)

10/24 – 01/25 **Chinese Language Scholar** *Shanghai Jiao Tong University, Shanghai, People's Republic of China* – (Incoming)
- Full-time semester program for Chinese language and culture

10/23 – 09/24 **Master Thesis Project** *National University of Singapore, Centre for Quantum Technologies & Yale-NUS College*
- Quantum Optimal Control with Reinforcement Learning (supervised by Hui Khoon Ng & Martin Plenio)

09/22 – 06/23 **Exchange Year** *National Taiwan University, Taipei, Republic of China* – (GPA: 4.0)
- Focus: Quantum Computing, High-Performance Computing and Machine Learning & AI

04/21 – 09/24 **MSc Physics and Management & MSc Physics (Double Degree)** *Ulm University, Ulm, Germany* – (current GPA: 3.9)
- Focus: Quantum Computing, Financial Modelling, Business Analytics (projects in R & Python)

10/17 – 03/21 **BSc Physics and Management** *Ulm University, Ulm, Germany*
- Focus: Statistical Mechanics, Business Analytics; Thesis on NV Centre dynamics (Matlab: grade: 3.7)

Coding Projects

01/24 – 06/24 **Multivariate Bicycle Codes** *Research Project with A*STAR Singapore & TII Abu Dhabi* – Link to publication
- Introduced new class of Quantum Error Correction codes improving encoding efficiency (Python – OOP)

10/23 – 09/24 **Context-Aware Gate Calibration with Deep Reinforcement Learning** *Research Project at CQT* – Link to GitHub
- Use Proximal Policy Optimization to design custom quantum control pulses (PyTorch, Gym, JAX – OOP)

02/23 – 06/23 **Study on Deep Learning Training with NVIDIA GPU** *HPC & AI class final project at National Taiwan University*
- Benchmark data and model parallelization on a GPU against CPU-based approaches (Python, CUDA)

Leadership & Achievements

12/23 – today **Organizer** *Singapore Quantum Technologies Meetup Group*
- Organizing regular sessions for quantum enthusiasts hosting speakers from industry & academia

09/22 – 06/23 **Baden-Württemberg Scholar & Regional Lead Taiwan** *Baden-Württemberg Foundation*
- Responsible for organizing network and cultural activities among scholarship holders in Taiwan
- Raised >15,000 EUR funding for my academic stays in Singapore and Taiwan

08/21 – today **President** *German Entrepreneur Prize for Students Alumni Association*
- Leading a team of four, overseeing the budget, planning networking and career development events

06/16 **2nd in Nationwide Business Plan Competition** *German Entrepreneur Prize for Students*

Skills & Interests

Languages	German (native), English (full working proficiency), Mandarin Chinese (est. HSK 4), French (est. B1)
Coding	Python (proficient); R, Matlab (intermediate); SQL, Bash, Git (basic)
Technologies	MS Office suite, LaTeX, Cloud Computing (AWS, Google Cloud), HPC Clusters
Interests	Swimming, tennis, cooking haute-cuisine, language tandems